

Generation Interconnection System Impact Study Report

For Queue Project Q21-1881

Big Sandy Peaker Plant 138 kV

December 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1881. The Interconnection Customer proposes a 105.4 MW solar photovoltaic facility interconnecting at the Big Sandy Peaker Plant 138 kV substation in the Kentucky area. The request was received on September 16, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 607 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,103,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Big Sandy Peaker Plant to N Findlay 138 kV	Big Sandy Peaker Plant to Ceredo Generating Sta Substation 138 kV	111.3	432	100.0
Big Sandy Peaker Plant to Ceredo Generating Sta 138 kV	Flewellen to North Findlay 138 kV	109.8	538	100.0
Baker to Flewellen 138 kV Line	Big Sandy Peaker Plant to Ceredo Generating Station 138 kV	106.2	547	100.0
Fewellen to N Findlay 138 kV	Big Sandy Peaker Plant to Ceredo Generating Station 138 kV L	105.9	557	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Sandy Peaker Plat to N Findlay 138 kV line	Rebuild line with higher-capacity conductor	\$3,139,000	21% (\$669,000)
Big Sandy Peaker Plant to Ceredo Generating Sta 138 kV	Reconductor line	\$2,290,000	17% (\$388,000)
Baker to Flewellen 138 kV Line	Reconductor line	\$3,715,000	17% (\$630,000)
N Findlay to Flewellen 138 kV	Reconductor line	\$2,452,000	17% (\$416,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$2,103,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.